

The artifact is my CS-360 Android weight tracking application, which I originally made during my CS-360 course. For Enhancement Two, I focused on the Algorithms and Data Structures category. I added a new `WeightRecordSorter` Java class and one line in `DataDisplayActivity.java` that calls `WeightRecordSorter.sortChronologically(weightRecords)`. The new class sorts weight records by date from oldest to newest instead of relying on the order returned by the database.

I selected this artifact because it helps show my ability to improve an existing application by applying a better algorithm and an appropriate data structure. The new sorter works with the existing `List<WeightRecord>` and uses comparisons with other data and organizes the records chronologically. The algorithm is a bubble sort, with a worst case complexity of $O(n^2)$ when records arrive newest first and a best case complexity of $O(n)$ when they are already ordered. Since SQLite was already returning the records in insertion order, many real world cases should be closer to the best case. I also kept the sorting logic in its own `WeightRecordSorter` class rather than placing it directly in the activity, making the code more readable and testable. Dates are parsed using the existing yyyy-mm-dd format, and invalid dates are moved to the end rather than causing the application to crash.

While I had already implemented my `WeightRecord` object and switched away from parallel `ArrayLists` that used strings and integers in my previous enhancement to make it work, I am actually building on that change here and using it to help apply an algorithm to my dataset and overall clean up how my data is stored/organized. That part of enhancement 1 was meant to be included here, but I had to make that change in Enhancement 1 for it to function properly.

I believe this enhancement supports Course Outcomes 3 and 4. It shows my ability to apply a better algorithm to organize data and improve the functionality of an existing application. It also shows me

using android studio, java, and SQLite, with a structured solution to improve my user experience. My overall outcome coverage plan has not changed from module 1, and I still plan to use the remaining enhancements to demonstrate additional skills in software engineering and database development.

This enhancement taught me more about choosing and implementing an algorithm based on the way data is actually stored and returned by an application. I also learned that separating the sorting logic into its own class can make the code easier to understand and test. One challenge was making sure the date strings were parsed consistently and that invalid dates would not crash the application. Another challenge was understanding the difference between bubble sort and other sorting approaches, the final implementation uses adjacent comparisons and swaps, repeating passes until no exchanges are needed.